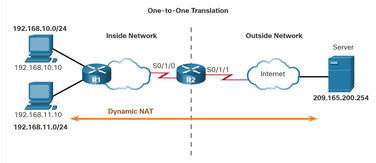
Dynamic NAT

6.5.1

Dynamic NAT Scenario

In this topic, you will learn how to configure and verify dynamic NAT. It includes a Packet Tracer activity to test your skills and knowledge. Although static NAT provides a permanent mapping between an inside local address and an inside global address, dynamic NAT automatically maps inside local addresses to inside global addresses. These inside global addresses are typically public IPv4 addresses. Dynamic NAT, like static NAT, requires the configuration of the inside and outside interfaces participating in NAT with the **ip nat inside** and **ip nat outside** interface configuration commands. However, where static NAT creates a permanent mapping to a single address, dynamic NAT uses a pool of addresses.

The example topology shown in the figure has an inside network using addresses from the RFC 1918 private address space. Attached to router R1 are two LANs, 192.168.10.0/24 and 192.168.11.0/24. Router R2, the border router, is configured for dynamic NAT using a pool of public IPv4 addresses 209.165.200.226 through 209.165.200.240.



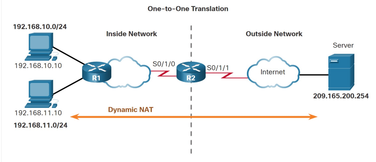
The pool of public IPv4 addresses (inside global address pool) is available to any device on the inside network on a first-come first-served basis. With dynamic NAT, a single inside address is translated to a single outside address. With this type of translation there must be enough addresses in the pool to accommodate all the inside devices needing concurrent access to the outside network. If all addresses in the pool are in use, a device must wait for an available address before it can access the outside network.

Note: Translating between public and private IPv4 addresses is by far the most common use of NAT. However, NAT translations can occur between pair of IPv4 addresses.

6.5.2

Configure Dynamic NAT

The figure shows an example topology where the NAT configuration allows translation for all hosts on the 192.168.0.0/16 network. This includes the 192.168.10.0 and 192.168.11.0 LANs when the hosts generate traffic that enters interface S0/1/0 and exits S0/1/1. The host inside local addresses are translated to an available pool address in the range of 209.165.200.226 to209.165.200.240.



Click each button for a description and example of each step to configure static NAT.

Step 1 | Step 2 | Step 3 | Step 4 | Step 5

Step 1

Define the pool of addresses that will be used for translation using the **ip nat pool** command. This pool of addresses is typically a group of public addresses. The addresses are defined by indicating the starting IPv4 address and the ending IPv4 address of the pool. The **netmask** or **prefix-length** keyword indicates which address bits belong to the network and which bits belong to the host for that range of addresses.

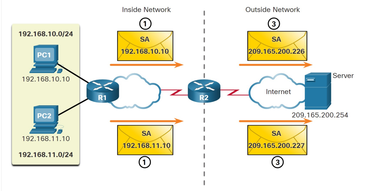
In the scenario, define a pool of public IPv4 addresses under the pool name NAT-POOL1.

```
R2(config)# ip nat pool NAT-POOL1 209.165.200.226 209.165.200.240 netmask 255.255.255.224
```

Analyze Dynamic NAT - Inside to Outside

Using the previous configuration, the next two figures illustrate the dynamic NAT translation process between two clients and the web server.

The figure below is used to illustrate the traffic flow from the inside network to the outside.



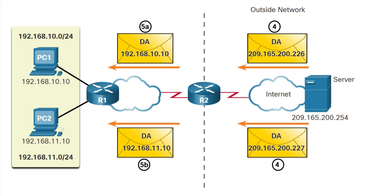
IPv4 NAT Pool	
Inside Local Address Pool	**Inside Global Address**
192.168.10.10	209.165.200.226
192.168.11.10	209.165.200.227

1. The hosts with the source IPv4 addresses of 192.168.10.10 (PC1) and 192.168.11.10 (PC2) send packets requesting a connection to the server at the public IPv4 address 209.165.200.254.
2. R2 receives the first packet from host 192.168.10.10. Because this packet was received on an interface configured as an inside NAT interface, R2 checks the NAT configuration to determine if this packet should be translated. The ACL permits this packet, so R2 will translate the packet. R2 checks its NAT table. Because there is no current translation entry for this IPv4 address, R2 determines that the source address 192.168.10.10 must be translated. R2 selects an available global address from the dynamic address pool and creates a translation entry, 209.165.200.226. The original source IPv4 address 192.168.10.10 is the inside local address and the translated address is the inside global address 209.165.200.226 in the NAT table. For the second host, 192.168.11.10, R2 repeats the procedure, selects the next available global address from the dynamic address pool, and creates a second translation entry, 209.165.200.227.
3. R2 replaces the inside local source address of PC1, 192.168.10.10, with the translated inside global address of 209.165.200.226 and forwards the packet. The same process occurs for the packet from PC2 using the translated address of 209.165.200.227.

6.5.4

Analyze Dynamic NAT - Outside to Inside

The figure below illustrates the remainder of the traffic flow between the clients and the server from the outside to the inside direction.



IPv4 NAT Pool	
Inside Local Address Pool	**Inside Global Address**
192.168.10.10	209.165.200.226
192.168.11.10	209.165.200.227

4. The server receives the packet from PC1 and responds using the IPv4 destination address of 209.165.200.226. When the server receives the second packet, it responds to PC2 using the IPv4 destination address of 209.165.200.227.
5. (a) When R2 receives the packet with the destination IPv4 address of 209.165.200.226, it performs a NAT table lookup. Using the mapping from the table, R2 translates the address back to the inside local address 192.168.10.10 and forwards the packet toward PC1.
   (b) When R2 receives the packet with the destination IPv4 address of 209.165.200.227, it performs a NAT table lookup. Using the mapping from the table, R2 translates the address back to the inside local address 192.168.11.10 and forwards the packet toward PC2.
6. PC1 at 192.168.10.10 and PC2 at 192.168.11.10 receive the packets and continue the conversation. The router performs Steps 2 to 5 for each packet. (Step 6 is not shown in the figures.)

6.5.5

Verify Dynamic NAT

The output of the **show ip nat translations** command displays all static translations that have been configured and any dynamic translations that have been created by traffic.

```
R2# show ip nat translations
Pro Inside global      Inside local       Outside local      Outside global
--- 209.165.200.228    192.168.10.10      ---                ---
--- 209.165.200.229    192.168.11.50      ---                ---
R2#
```

Adding the **verbose** keyword displays additional information about each translation, including how long ago the entry was created and used.

```
R2# show ip nat translation verbose
Pro Inside global      Inside local       Outside local      Outside global
tcp 209.165.200.228    192.168.10.10      ---                ---
    create 00:02:11, use 00:02:11 timeout:86400000, left 23:57:49, Map-Id(In): 1,
    flags:
none, use_count: 0, entry-id: 10, in_entries: 0
tcp 209.165.200.229    192.168.11.50      ---                ---
    create 00:02:10, use 00:02:10 timeout:86400000, left 23:57:49, Map-Id(In): 1,
    flags:
none, use_count: 0, entry-id: 12, in_entries: 0
R2#
```

By default, translation entries time out after 24 hours, unless the timers have been reconfigured with the **ip nat translation timeout** *timeout-seconds* command in global configuration mode.

To clear dynamic entries before the timeout has expired, use the **clear ip nat translation** privileged EXEC mode command as shown.

```
R2# clear ip nat translation *
R2# show ip nat translation
```

It is useful to clear the dynamic entries when testing the NAT configuration. The **clear ip nat translation** command can be used with keywords and variables to control which entries are cleared, as shown in the table. Specific entries can be cleared to avoid disrupting active sessions. Use the **clear ip nat translation** * privileged EXEC command to clear all translations from the table.

Command	Description
`clear ip nat translation *`	Clears all dynamic address translation entries from the NAT translation table.
`clear ip nat translation inside`*global-ip local-ip* `outside` *local-ip global-ip*	Clears a simple dynamic translation entry containing an inside translation or both inside and outside translation.
`clear ip nat translation protocol inside`*global-ip global-port local-ip local-port* `outside`*local-ip local-port global-ip global-port*	Clears an extended dynamic translation entry.

Note: Only the dynamic translations are cleared from the table. Static translations cannot be cleared from the translation table.

Another useful command, **show ip nat statistics**, displays information about the total number of active translations, NAT configuration parameters, the number of addresses in the pool, and how many of the addresses have been allocated.

```
R2# show ip nat statistics
Total active translations: 4 (0 static, 4 dynamic; 0 extended)
Peak translations: 4, occurred 00:31:43 ago
Outside interfaces:
  Serial0/1/1
Inside interfaces:
  Serial0/1/0
Hits: 47  Misses: 0
CEF Translated packets: 47, CEF Punted packets: 0
Expired translations: 5
Dynamic mappings:
-- Inside Source
[Id: 1] access-list 1 pool NAT-POOL1 refcount 4
 pool NAT-POOL1: netmask 255.255.255.224
        start 209.165.200.226 end 209.165.200.240
        type generic, total addresses 15, allocated 2 (13%), misses 0
(output omitted)
R2#
```

Alternatively, you can use the **show running-config** command and look for NAT, ACL, interface, or pool commands with the required values. Examine these carefully and correct any errors discovered. The example shows the NAT pool configuration.

```
R2# show running-config | include NAT
ip nat pool NAT-POOL1 209.165.200.226 209.165.200.240 netmask 255.255.255.224
ip nat inside source list 1 pool NAT-POOL1
```

6.5.6

Packet Tracer - Configure Dynamic NAT

In this Packet Tracer, you will complete the following objectives:

- Configure Dynamic NAT
- Verify NAT Implementation

Configure Dynamic NAT

Configure Dynamic NAT